RESEARCH PRESENTATION

RELIO

Concept drift detection and drift-type identification with optimal transport

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Context and motivation

Streams change

Data distributions shift over time (concept drift). Models trained on past data degrade.

Reactive monitoring is late

Watching model performance only reveals drift after accuracy has already dropped.

Drift needs more than a flag

Few methods say how drift appeared, yet its type affects how to adapt.

Problem statement

How can drift in a data stream be detected by comparing distributions directly, and its type identified, without waiting for model performance to drop?

Detect

When P(X, y) changes between windows

Characterize

Sudden, incremental, gradual or recurrent

Support

Multidimensional data; supervised or unsupervised models

Research objectives

1 · Detect early

Monitor underlying distributions with optimal transport (OT) instead of model error.

2 · Grade severity

Two thresholds: alert (minor) and detection (significant).

3 · Name the type

Classify drift to guide the adaptation strategy.

4 · Validate

Test on synthetic and real streams and compare with EMD and JS.

Related work and research gap

Work	Contribution	Limitation noted in the paper
Zhao & Koh [15]	Sliding-window Wasserstein drift detection, plus feature importance	Detects drift; no drift-type identification
OTDD, Laha & Verma [16]	Wasserstein barycenters, KL divergence; targets labeling cost	Supervised setting; detection only
HDDDM	Histograms + Hellinger distance per variable	Needs immediate labels
ADAPT [17]	OT-based dissimilarity for customs fraud	Application-specific

Gap: drift-type identification on top of OT-based detection.

Approach: compare windows with OT

Reference window → Current window → OT distance → Compare to thresholds

Cost functions

Euclidean, standardized Euclidean, Mahalanobis

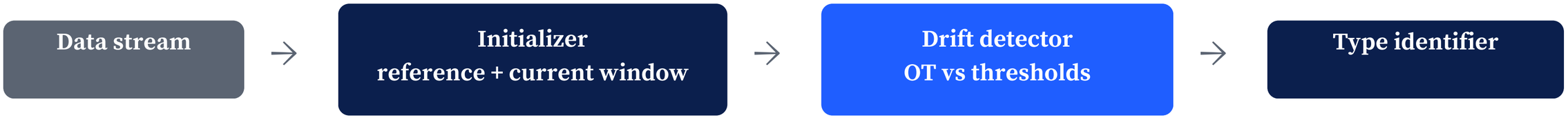
Thresholds

Set as % above a baseline distance between two random samples of the data

Windows

Sliding windows, user-defined size (logical or physical time)

RELIO architecture



No change

Below alert. Reference updated with a random element of the current window.

Minor change

Between alert and detection. Flags that partial adjustment is needed.

Significant change

Above detection. Flags adaptation; current window becomes the new reference.

Drift-type identification

Sudden

Detection occurs within the sudden-drift threshold after onset (default: 1 window).

Recurrent

New concept matches an earlier stored concept (OT below alert threshold).

Incremental

Distance non-decreasing over at least 3 windows up to detection.

Gradual

At least 5 alternating concept durations: source 1 shrinking, source 2 growing.

Applied only after the new concept stays stable for the stability threshold (default 3 windows). Order checked in code: sudden, recurrent, incremental, gradual.

Experimental setup and datasets

Dataset	Kind	Size	Role
Iris-based simulation	Synthetic	340 points (gradual stream)	Gradual, sudden, incremental, recurrent
Insects Sudden	Synthetic benchmark	24,150 × 33	Sudden drift near index 2000
Ozone	Real	2,534 × 72	Recurrent distributions
Asfault	Real	8,563 × 64	Several changes

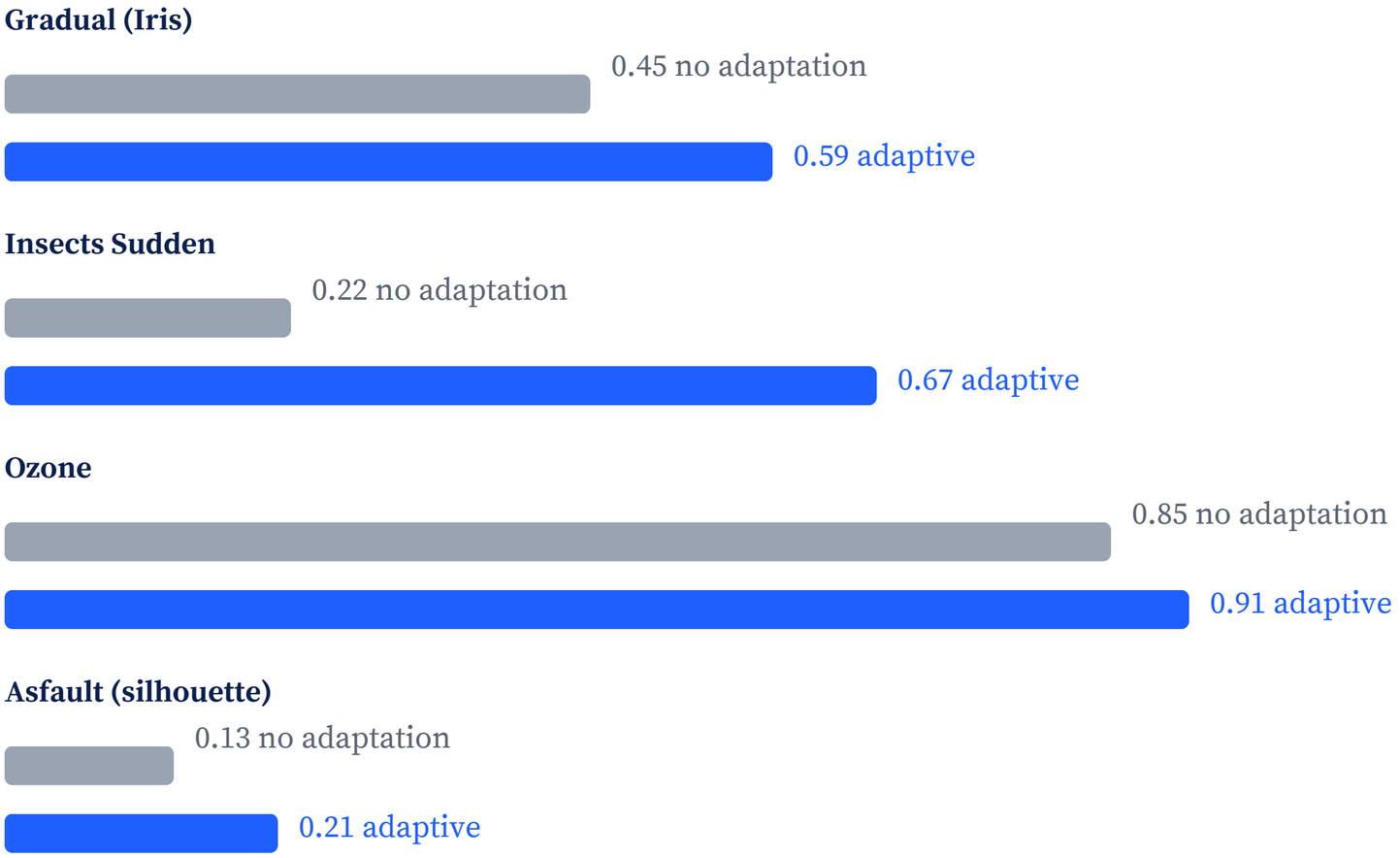
Baselines: EMD and JS detectors. Impact: adaptive vs non-adaptive SGD classifiers (Asfault: two clustering models, silhouette score).

Results: detection and type

Dataset	Detection vs EMD / JS	RELIO type
Simulated gradual	All detected drifts, same positions	Gradual
Insects Sudden	Positions differ among methods; all detect from index 2000	Sudden
Ozone	Same positions	Recurrent
Asfault	Most drifts found; EMD missed index 3500; RELIO alert at 1000	Incremental shown in Fig. 25 (blank in Table 1)

Insects: RELIO missed the change EMD and JS flagged at 600, and JS missed 1400. Detection quality is described, not scored.

Results: effect of adapting



Paper Table 1 · single runs; metric values as reported

Limitations and conclusion

Limitations (our reading)

No detection delay or false-alarm metrics. Single runs, hand-set thresholds (10% to 70%). Simulations built from Iris. Early detection before degradation is not measured.

Implemented

OT detection, alert/detection thresholds, four drift types, EMD/JS comparison.

Future work

Type-specific adaptation, better thresholds, more OT metrics and costs, feature-level drift impact.